

STAT400: Midterm Review

Fall 2025

Sampling

Inverse CDF

- 1) • I think it would also be helpful to review how to find an inverse CDF, since I'm still a bit unsure about that.

- 2) • Could we also go over approaches for finding the c constant for the envelope in the accept-reject method?

- 3) • Lastly, going over the accept-reject method and Monte Carlo sampling would be great!

- 4) • What are some ways to come up with an envelope when the uniform does not work?

- 5) • What are some good ways to prove that the envelope is valid?

- 6) • I am really confused about the method of using envelopes. I don't really understand how to use them, nor why or when we would want to use them.

- 7) • I would like you to cover how to create the best 'envelope' for accept-reject method. And please provide an example, I noticed in our notes we don't really have a defined envelope in the examples provided; I just want to see the thinking process on how to create the envelope.

- 8) • I would like to request that we go over envelope functions a bit more, and I am struggling a bit to understand specifically what they are and how they are found.

Accept-Reject

1) Inverse CDF:

Continuous

1. Derive the inverse cdf

To do this, let $F(x) = u$ and solve for x to find $x = F^{-1}(u)$.

2. Write a function to compute $F^{-1}(u)$ (in R)

3. For $i = 1, \dots, M$

a) Sample $u_i \sim \text{Unif}(0,1)$

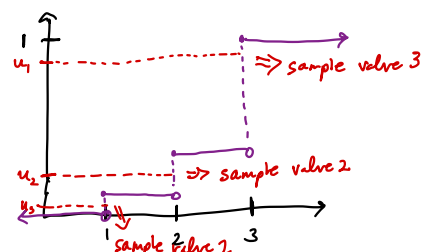
b) compute $x_i = F^{-1}(u_i)$

Discrete

1. Generate $u \sim \text{Unif}(0,1)$

2. Find x_i where $F_X(x_{i-1}) < u \leq F_X(x_i)$

e.g. $X \in \{1, 2, 3\}$, $P(X=1) = 0.1$, $P(X=2) = 0.3$, $P(X=3) = 0.7$



2) - 8)

Accept - Reject

Algorithm:

1. Find suitable g and c

g needs to be ① easy to sample from, ② easy to evaluate, and ③ support of g includes support of f .

$c \cdot g(x) \geq f(x) \forall x$

2. Sample $Y \sim g$

3. Sample $U \sim \text{Unif}(0,1)$

4. If $U < f(Y)/c \cdot g(Y)$, accept Y

Repeat 2-4 until you have enough samples.

If support is finite, can always choose $\text{Unif}(a,b)$ where $[a,b] = \text{support of } f$.

If support is not finite:

Do not pick uniform!

① Pick g from the table of dsns

→ plot f to see if there are asymptotes in $\mathbb{R}$. If so, try to pick g w/ same asymptote.

→ pay special attention to $\mathbb{R}$ and support of g

② Find c so that $c \cdot g(x) \geq f(x) \forall x$

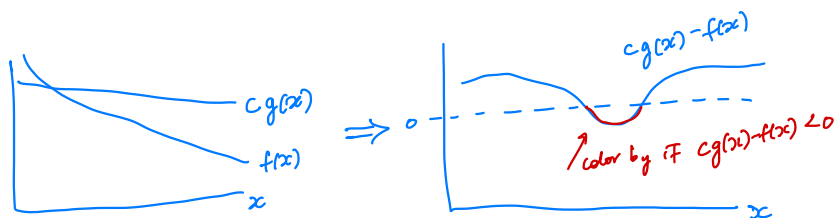
> try to solve for c analytically

> make many plots over a wide range of x 's

① pick many c 's

② pick many x 's

③ For each c , plot



Accepted by c

Example: let's use accept-reject to generate $Y \sim N(0,1)$. let $g = \text{Double exponential}(0,1)$, i.e. $g(x) = \frac{1}{2} \exp(-|x|)$, $x \in \mathbb{R}$.

$$\Rightarrow f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

Want to find a constant c s.t.

$$c \geq \frac{f(x)}{g(x)} \quad \forall x \in \mathbb{R}$$

Not make some plots

$$\text{Turns out } \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2} + |x|\right) \leq \frac{2}{\sqrt{2\pi}} \exp\left(\frac{1}{2}\right) = c.$$

Finding c :

$$\frac{f(x)}{g(x)} = \frac{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)}{\frac{1}{2} \exp(-|x|)} = \frac{2}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2} + |x|\right) \leq c$$

$$\text{Note } \frac{f(Y)}{c \cdot g(Y)} = \frac{f(Y)}{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{Y^2}{2}\right)} = \exp\left(-\frac{1}{2}(Y^2 + 2|Y| - 1)\right) = \exp\left(-\frac{1}{2}(|Y| - 1)^2\right)$$

⇒ Algorithm:

① Sample $Y \sim \text{Double Exp}(0,1)$

② Sample $U \sim \text{Unif}(0,1)$

③ If $U < f(Y)/c \cdot g(Y)$ accept.

How to sample??

Inverse CDF!

$$G(x) = \int_{-\infty}^x \frac{1}{2} \exp(-|t|) dt = \begin{cases} \int_{-\infty}^x \frac{1}{2} \exp(t) dt = \frac{1}{2} \exp(x) - \frac{1}{2} \exp(-\infty) & \text{if } x \leq 0 \\ \int_{-\infty}^x \frac{1}{2} \exp(-t) dt = \frac{1}{2} + \int_0^x \frac{1}{2} \exp(-t) dt & \text{if } x > 0 \end{cases}$$

$$\Rightarrow \text{Set } U = G(x) \quad a) \quad u = \frac{1}{2} \exp(x) \quad \text{when } x \leq 0, \text{ i.e. when } u \leq \frac{1}{2}$$

$$\log 2u = x = G^{-1}(u)$$

$$b) \quad u = 1 - \frac{1}{2} \exp(-x) \quad \text{when } x > 0, \text{ i.e. when } u > \frac{1}{2}$$

$$-\log(2(1-u)) = x \quad \text{when } x > 0, \text{ i.e. when } u > \frac{1}{2}$$

$$= G^{-1}(u)$$

$$-x < 0$$

$$\exp(-x) < \exp(0) = 1$$

$$| -\frac{1}{2} \exp(-x) > | -\frac{1}{2} \Rightarrow u > \frac{1}{2}$$

$\Rightarrow$ To sample $y \sim \text{Double Exp}(0,1)$,
 ① Draw $u \sim \text{Unif}(0,1)$
 ② $x = G^{-1}(u) = \begin{cases} \log 2u & \text{when } u \leq \frac{1}{2} \\ -\log(2(1-u)) & \text{when } u > \frac{1}{2} \end{cases}$

Monte Carlo Integration

- ① Could we go over a specific example and/or a review of finding the $\text{Var}(g(x))$ using a Monte Carlo approach?
- ② I was wondering if, when we are using methods 1 & 2 for Monte Carlo Integration, is the uniform distribution the only distribution we can use? In other words, if the uniform distribution isn't a good option, is method 3 our go-to? If not, how would we use other distributions in that way? *Case by case. Really, whatever works (mathematically).*
- ③ In our review on Tuesday, would you elaborate more on why Monte Carlo integration is superior to approximating an integral via the sum of polygons? I know that we talked about how the estimate produced by Monte Carlo converges towards the true value of the integral as m increases, but I don't quite understand why this is better than increasing the number of polygons, especially because Monte Carlo inherently involves randomness. *Appealing feature is that we use the properties of the sample mean as an estimator of the pop. mean $\Rightarrow$ interval estimation! Also randomness allows us to explore a high dimensional space without having to do systematic search, can be faster.*

① Example:

Imagine want to estimate $\theta = \int_2^3 e^{-x^3} dx$.

let $f \sim \text{Unif}(2,3) \Rightarrow f(x) = \begin{cases} 1 & x \in (2,3) \\ 0 & \text{o.w.} \end{cases}$

$$\theta = \int_2^3 \underbrace{e^{-x^3}}_{g(x)} \cdot 1 dx = E[\bar{e}^{X^3}], \quad X \sim \text{Unif}(2,3)$$

Then to estimate θ ,

① Sample $X_1, \dots, X_m \sim \text{Unif}(2,3)$

② Compute $\hat{\theta} = \frac{1}{m} \sum_{i=1}^m \bar{e}^{X_i^3}$

How good is our estimate? Bias = 0 ✓ What about variance?

$$\text{Var}[\hat{\theta}] = \text{Var}\left[\frac{1}{m} \sum_{i=1}^m g(X_i)\right] \stackrel{\text{iid}}{=} \frac{1}{m^2} \sum_{i=1}^m \text{Var}[g(X_i)] = \frac{1}{m} \boxed{\text{Var}[g(X_i)]}$$

don't know this!

$$\text{Var}[g(X_i)] = E[(g(X_i) - E[g(X_i)])^2]$$

$$= E[(\bar{e}^{X_i^3} - \theta)^2], \quad X_i \sim \text{Unif}(2,3)$$

estimate w/ sample mean!

$$\hat{\text{Var}} g(X_i) = \frac{1}{m} \sum_{i=1}^m (\bar{e}^{X_i^3} - \hat{\theta})^2$$

or estimate variance w/ sample variance!

$$\hat{\text{Var}} g(X) = \frac{1}{m-1} \sum_{i=1}^m (\bar{e}^{X_i^3} - \hat{\theta})^2$$

$$\Rightarrow \hat{\text{Var}}(\hat{\theta}) = \frac{1}{m} \left(\frac{1}{m} \sum_{i=1}^m (\bar{e}^{X_i^3} - \hat{\theta})^2 \right)$$

Question:
 Can we use the same $X_1, \dots, X_m \sim \text{Unif}(2,3)$ from step ①?

Other Questions

- 1) • Also, from Homework 3, we had to do integration by parts; could we go over a similar example during Tuesday's review?
- 2) • Going over marginal densities and trying a new example that isn't the drive-up and walk-up window one used from notes.
- 3) • Could we review some examples using group_by, summarise, filter (etc.) like we did earlier in the semester? *Do some code examples.*
- 4) • I was wondering if it would be possible to elaborate on what we can expect from the coding-based questions on the exam.
 - ↳ looking at code - finding errors or telling me what output would be
 - finding a particular line of code that does something.
 - ↳ writing your pseudocode (plan).

1) ex. $\int x e^{6x} dx$

IBP: $\int u dv = uv - \int v du$

take derivatives $\downarrow$ take integrals $\downarrow$

+	x	e^{6x}
-	1	$\frac{1}{6} e^{6x}$
+	0	$\frac{1}{36} e^{6x}$

make sure this side eventually gets to 0!

$$= \frac{x}{6} e^{6x} - \frac{1}{36} e^{6x}$$

vs. $u = x \Rightarrow du = dx$

$$dv = e^{6x} dx \Rightarrow v = \int dv = \int e^{6x} dx = \frac{1}{6} e^{6x}$$

$$= x \cdot \frac{1}{6} e^{6x} - \int \frac{1}{6} e^{6x} dx$$

$$= \frac{x}{6} e^{6x} - \frac{1}{36} e^{6x} \checkmark$$

2) Define a joint pdf by $f(x, y) = \begin{cases} 6xy^2 & 0 < x < 1 \text{ and } 0 < y < 1 \\ 0 & \text{o.w.} \end{cases}$

- check that $f(x, y)$ is a pdf

$$f(x, y) \geq 0 \quad \forall \quad x, y \in \mathbb{R}^2 \checkmark$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = \int_0^1 \int_0^1 6xy^2 dx dy = \int_0^1 \left[3x^2 y^2 \right]_{x=0}^1 dy = \int_0^1 3y^2 dy = \left[y^3 \right]_0^1 = 1 \checkmark$$

- get marginals

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 6xy^2 dy = \left[2xy^3 \right]_{y=0}^1 = 2x, \quad 0 < x < 1$$

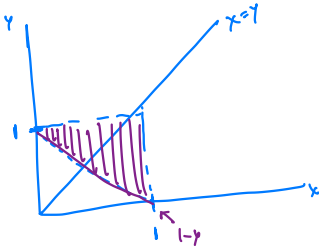
$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 6xy^2 dx = \left[3x^2 y^2 \right]_{x=0}^1 = 3y^2, \quad 0 < y < 1.$$

- Use marginals:

$$P\left(\frac{1}{2} < X < \frac{3}{4}\right) = \int_{\frac{1}{2}}^{\frac{3}{4}} f_X(x) dx = \int_{\frac{1}{2}}^{\frac{3}{4}} 2x dx = x^2 \Big|_{\frac{1}{2}}^{\frac{3}{4}} = \left(\frac{3}{4}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{5}{16}$$

- Use joint:

$$P(X+Y \geq 1)$$



when $x+y \geq 1, 0 < x < 1, 0 < y < 1 \Rightarrow$
 $x \geq 1-y, 0 < x < 1, 0 < y < 1 \Rightarrow$
 $1-y \leq x < 1, 0 < y < 1$

$$3(1-2y+y^2)y^2$$

$$\downarrow = 3 - 6y^3 + 3y^4$$

$$\text{So } P(X+Y \geq 1) = \int_0^1 \int_{1-y}^1 f(x,y) dx dy = \int_0^1 \int_{1-y}^1 6xy^2 dx dy = \int_0^1 \left[3x^2 y^2 \right]_{1-y}^1 dy = \int_0^1 (3y^2 - 3(1-y)^2 y^2) dy =$$

$$\left[y^3 - y^3 + \frac{6}{4} y^4 - \frac{3}{5} y^5 \right]_0^1 = \frac{6}{4} - \frac{3}{5} = \frac{9}{10}$$

Note to self:
Don't do
if don't
have to